

ESE 5700: Digital Integrated Circuits and VLSI Fundamentals

Project 1, Part 1: The 6T Bitcell

Tyrone Marhguy

Victor Wanjohi

University of Pennsylvania
Due: 9/29/2026



Team and contributions

Member	Contributions (this part)
Tyrone Marhguy	<i>TBD</i>
Victor Wanjohi	<i>TBD</i>

Both team members are responsible for understanding the full design and report.

1. Sizing (β and γ)

Hand analysis

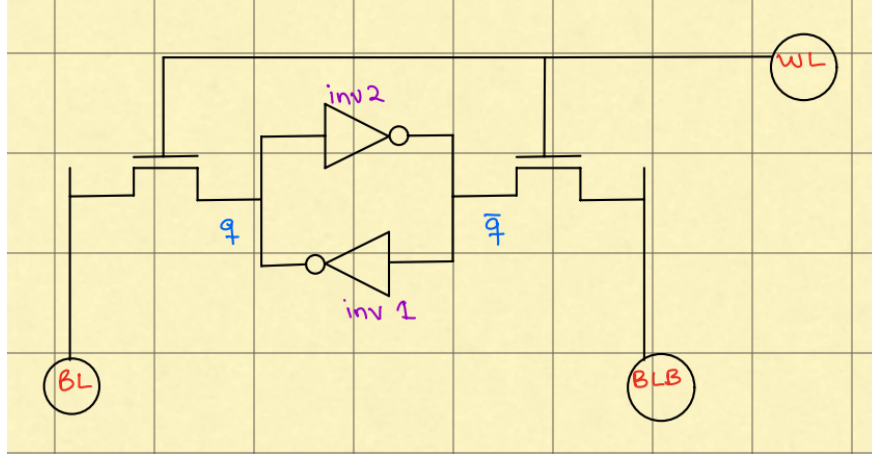


Figure 1: 6T cell read-disturb setup: $Q = 0$ stored, wordline asserted, bitlines precharged (hand sketch).

Read stability

Assume the stored state $Q = 0$ ($\bar{Q} \approx V_{DD}$) for read-stability analysis. With $Q = 0$ the “0” node is pulled down by the NMOS inside INV1 while the bitline (precharged to V_{DD}) drives current through the pass-gate, so the disturbed node follows an approximate voltage-divider model:

$$\frac{V_Q}{V_{DD}} = \frac{R_{PD1}}{R_{PG} + R_{PD1}} = \frac{1}{1 + R_{PG}/R_{PD1}}. \quad (1)$$

From transistor physics $R_T \propto 1/W_T$ at fixed L . Defining the cell ratio $\beta = (W/L)_{PD}/(W/L)_{PG} = W_{PD}/W_{PG}$ gives $\beta \approx R_{PG}/R_{PD}$, hence

$$\frac{V_Q}{V_{DD}} \approx \frac{1}{1 + \beta}, \quad V_Q \approx \frac{V_{DD}}{1 + \beta} : \quad (2)$$

a larger β lowers the read-disturb bump. Whether that bump flips the cell is set by the inverter switching voltage V_M ($V_{in} = V_{out}$), where both devices conduct.

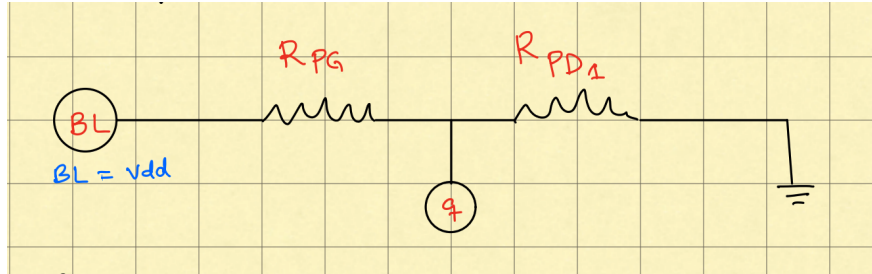


Figure 2: Approximate resistive divider for the read disturb: precharged bitline through R_{PG} against the pull-down R_{PD1} to ground (hand sketch).

Current balance at the equilibrium disturb voltage: neglecting gate leakage, KCL gives $I_{PD} = I_{PG}$. The pass-gate has $D = G = V_{DD}$, $S = V_Q$, so $V_{DS} = V_{GS} = V_{DD} - V_Q$ and $V_{DS} > V_{GS} - V_{Tn}$ always: saturation, with $I_{PG} = \frac{1}{2}k_{PG}(V_{GS} - V_{Tn})^2$ and $k_{PG} = \mu_n C_{ox}(W/L)_{PG}$. The pull-down (pd1) has $V_D = V_Q$, $V_G \approx V_{DD}$, $V_S = 0$, so $V_{DS} = V_Q < V_{GS} - V_{Tn}$: triode, with $I_{PD} = k_{PD}[(V_{GS} - V_{Tn})V_{DS} - V_{DS}^2/2]$. Equating the two with $E = V_{DD} - V_{Tn}$ yields

$$V_Q^2(1 + \beta) + V_Q(-2E - 2E\beta) + E^2 = 0 \quad \Rightarrow \quad V_Q = E \left(1 \pm \sqrt{\frac{\beta}{1 + \beta}} \right), \quad (3)$$

of which the positive root exceeds $V_{DD} - V_{Tn}$ and is inconsistent with the assumed saturated pass-gate / triode pull-down, leaving

$$V_Q = (V_{DD} - V_{Tn}) \left(1 - \sqrt{\frac{\beta}{1 + \beta}} \right). \quad (4)$$

This assumes ideal long-channel square-law behavior, equal NMOS thresholds, and negligible body effect.

The disturbed node Q drives INV2, whose switching point satisfies $I_{DN} = |I_{DP}|$ with both devices in saturation:

$$V_M = \frac{V_{Tn} + B(V_{DD} - |V_{Tp}|)}{1 + B}, \quad B = \sqrt{\frac{k_P}{k_N}} = \sqrt{\frac{\gamma \mu_p}{\beta \mu_n}}, \quad (5)$$

with $k_P/k_N = \mu_p C_{ox}(W/L)_{PU}/(\mu_n C_{ox}(W/L)_{PG})$, $\beta = (W/L)_{PD}$, and $\gamma = (W/L)_{PG}$ as written in the hand-note derivation, so V_M depends on the supply, the threshold voltages, and the relative PMOS/NMOS strengths. Read stability requires $V_Q < V_M$:

$$(V_{DD} - V_{Tn}) \left(1 - \sqrt{\frac{\beta}{1 + \beta}} \right) < \frac{V_{Tn} + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}} (V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}}}. \quad (6)$$

Increasing β strengthens the pull-down relative to the pass-gate and lowers the disturb voltage, but β also moves V_M and costs area: the final cell ratio is not selected by minimizing V_Q alone.

Write stability

Assume the stored state $Q = 1$ ($\bar{Q} \approx 0$) for the write analysis, with a 0 written through BL. With BL driven to 0 V and WL at V_{DD} , the pass-gate pulls the “1” node down while the PMOS inside INV1, still on because $\bar{Q} \approx 0$, pulls it up to V_{DD} . The written node therefore follows an approximate voltage-divider model:

$$\frac{V_Q}{V_{DD}} = \frac{R_{PG}}{R_{PG} + R_{PU1}} = \frac{1}{1 + R_{PU1}/R_{PG}}. \quad (7)$$

From transistor physics $R_T \propto 1/(\mu_T W_T)$ at fixed L . The two devices are of opposite type, so the mobility ratio does not cancel as it does in the read divider. Defining the pull-up ratio $\gamma = (W/L)_{PU}/(W/L)_{PG} = W_{PU}/W_{PG}$ gives $R_{PU1}/R_{PG} \approx \mu_n/(\mu_p \gamma)$, hence

$$\frac{V_Q}{V_{DD}} \approx \frac{1}{1 + \frac{\mu_n}{\mu_p \gamma}}, \quad V_Q \approx V_{DD} \frac{\mu_p \gamma}{\mu_n + \mu_p \gamma} : \quad (8)$$

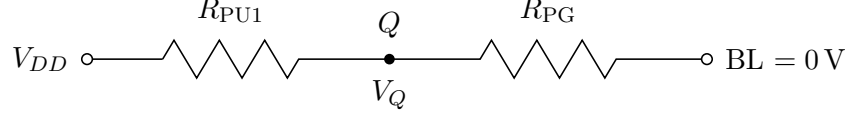


Figure 3: Approximate resistive divider for the write: supply through R_{PU1} against the pass-gate R_{PG} to the 0 V bitline.

a smaller γ lowers the written node. Whether it falls far enough to flip the cell is set by the switching voltage V_M ($V_{in} = V_{out}$) of INV2, which Q drives.

Current balance at the equilibrium write voltage: neglecting gate leakage, KCL gives $I_{PU} = I_{PG}$. The pass-gate has $G = V_{DD}$, $S = BL = 0$ and $D = Q$, so $V_{GS} = V_{DD}$ and $V_{DS} = V_Q < V_{GS} - V_{Tn}$ at the equilibrium point: triode, with $I_{PG} = k_{PG}[(V_{DD} - V_{Tn})V_Q - V_Q^2/2]$ and $k_{PG} = \mu_n C_{ox}(W/L)_{PG}$. The pull-up (pu1) has $S = V_{DD}$, $G = \bar{Q} \approx 0$ and $D = Q$, so $V_{SG} = V_{DD}$ and $V_{SD} = V_{DD} - V_Q > V_{SG} - |V_{Tp}|$ as long as $V_Q < |V_{Tp}|$: saturation, with $I_{PU} = \frac{1}{2}k_{PU}(V_{DD} - |V_{Tp}|)^2$ and $k_{PU} = \mu_p C_{ox}(W/L)_{PU}$. Equating the two with $E = V_{DD} - V_{Tn}$, $F = V_{DD} - |V_{Tp}|$ and $k_{PU}/k_{PG} = \gamma \mu_p/\mu_n$ yields

$$V_Q^2 - 2E V_Q + \gamma \frac{\mu_p}{\mu_n} F^2 = 0 \quad \Rightarrow \quad V_Q = E \pm \sqrt{E^2 - \gamma \frac{\mu_p}{\mu_n} F^2}, \quad (9)$$

of which the positive root exceeds $V_{DD} - V_{Tn}$ and is inconsistent with the assumed triode pass-gate, leaving

$$V_Q = (V_{DD} - V_{Tn}) - \sqrt{(V_{DD} - V_{Tn})^2 - \gamma \frac{\mu_p}{\mu_n} (V_{DD} - |V_{Tp}|)^2}. \quad (10)$$

This assumes ideal long-channel square-law behavior, a saturated pull-up ($V_Q < |V_{Tp}|$), an ideal 0 V bitline, and no help from PG2 pulling $\bar{Q}$ up.

The written node Q drives INV2, whose switching point V_M is the one in Eq. (5), with $B = \sqrt{\gamma \mu_p/(\beta \mu_n)}$. Write stability requires $V_Q < V_M$:

$$(V_{DD} - V_{Tn}) - \sqrt{(V_{DD} - V_{Tn})^2 - \gamma \frac{\mu_p}{\mu_n} (V_{DD} - |V_{Tp}|)^2} < \frac{V_{Tn} + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}} (V_{DD} - |V_{Tp}|)}{1 + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}}}. \quad (11)$$

Equation (10) holds only while the pull-up is saturated, $V_Q < |V_{Tp}|$. With the values used here V_M is about 0.37 V, above $|V_{Tp}| = 0.3$ V, so (11) cannot be solved for γ with (10) alone. Requiring the stricter $V_Q < V_{Tn}$ keeps (10) valid when $V_{Tn} \leq |V_{Tp}|$, guarantees that the NMOS of INV2 is off, and gives the required pull-up ratio in closed form:

$$\gamma < \frac{\mu_n}{\mu_p} \cdot \frac{V_{Tn}(2V_{DD} - 3V_{Tn})}{(V_{DD} - |V_{Tp}|)^2}. \quad (12)$$

At $V_{DD} = 0.8$ V, with the placeholder values $V_{Tn} = |V_{Tp}| = 0.3$ V and $\mu_n/\mu_p = 2$, (12) gives $\gamma < 1.68$. Evaluating (11) numerically at $\beta = 2$, with the triode pull-up current beyond $V_Q = |V_{Tp}|$, relaxes the limit to about $\gamma < 1.9$. For $\gamma = 1$ the divider (8) puts V_Q at 0.27 V and the current balance (10) at 0.15 V; the divider is pessimistic because a saturated pull-up supplies less current than a resistor would.

Table 1: Written-node voltage against pull-up ratio at $V_{DD} = 0.8\text{ V}$, $\mu_n/\mu_p = 2$.

γ	V_Q from (10) (V)	Margin to V_{Tn} (V)	Margin to V_M at $\beta = 2$ (V)
0.5	0.067	0.233	0.285
1	0.146	0.154	0.220
1.5	0.250	0.050	0.126
2	0.40, pull-up in triode	fails	fails

The bound scales directly with the NMOS-to-PMOS strength ratio. FinFET kits often have a PMOS much closer to the NMOS than the planar assumption of 2, and that decides whether a one-fin pass-gate can overpower a one-fin pull-up:

Table 2: Sensitivity of the write bound to the NMOS-to-PMOS strength ratio.

Strength ratio μ_n/μ_p	γ limit from (12)	V_Q at $\gamma = 1$ (V)	V_Q at $\gamma = 0.5$ (V)
2	1.68	0.146	0.067
1.5	1.26	0.211	0.092
1.25	1.05	0.276	0.113
1	0.84	fails	0.146

Reducing γ weakens the pull-up relative to the pass-gate and lowers the written node, but γ moves in whole-fin steps, a stronger pass-gate lowers β and loads the bitline, and γ also moves V_M : the final pull-up ratio is not selected by minimizing V_Q alone.

Chosen sizes

Device	(W/L)	Notes
Pull-down (PD)	<i>TBD</i>	
Pull-up (PU)	<i>TBD</i>	
Pass-gate (PG)	<i>TBD</i>	
β	<i>TBD</i>	
γ	<i>TBD</i>	

β sweep (schematic)

β	SNM_{read}	Notes
$1\times$	<i>TBD</i>	
$1.5\times$	<i>TBD</i>	
$2\times$	<i>TBD</i>	

Design point selected: *TBD*. Cell kept symmetric.

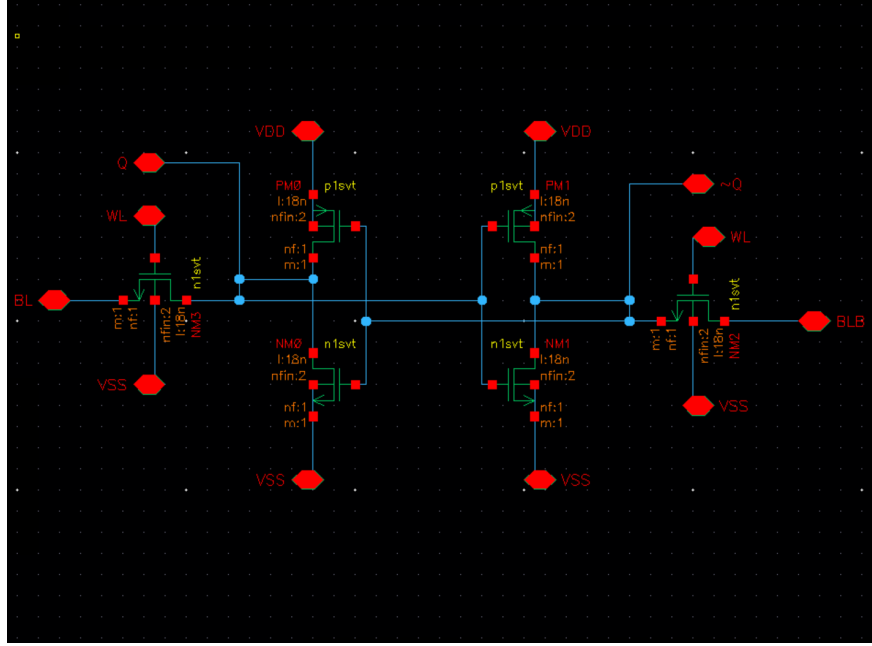


Figure 4: 6T bitcell schematic with transistor sizes (proj1_lib 6t-sram, Virtuoso capture).

2. Layout

DRC- and LVS-clean layout: wordline horizontal, bitline pair vertical, tileable for Part 2. Report area in μm^2 and F^2 ($F = 45\text{ nm}$).

Metric	Value
Cell area	<i>TBD</i> μm^2
Cell area in F^2	<i>TBD</i> F^2
DRC	<i>clean / summary</i>
LVS	<i>clean / summary</i>

3. Extraction

Post-layout parasitic extraction. Report per-cell bitline-node and wordline-node capacitances (loads for Part 2).

Capacitance	Value
C_{BL} (per cell)	<i>TBD</i>
C_{WL} (per cell)	<i>TBD</i>

4. Stability characterization (post-layout, $V_{DD} = 1.2\text{ V}$)

4a. Hold SNM

Butterfly plot with $WL = 0$; report $SNM_{\text{hold}} = \min(SNM_L, SNM_R)$.

4b. Read SNM

Butterfly plot with $WL = V_{DD}$, $BL=BLB= V_{DD}$; report SNM_{read} and DC voltage of the “0” node vs. the Section 1 estimate.

4c. Write margin

Store $Q = 1$, $WL = V_{DD}$, $BL = V_{DD}$; sweep BLB ; report $WM_{\text{BL}} = V_{\text{trip}}$.

4d. Cell current

Report I_{cell} from the “0”-side bitline ($WL = V_{DD}$, bitlines at V_{DD}).

4e. Supply dependence

V_{DD}	SNM_{read}	WM_{BL}
1.2 V	TBD	TBD
0.9 V	TBD	TBD
0.6 V	TBD	TBD

Still reads and writes at 0.6 V? *yes/no + notes.*

5. Functional check

Transient test with ideal WL and precharge pulses; bitline pair loaded by 15 additional per-cell bitline loads. Write 0, read; write 1, read. Show WL, BL, BLB, Q, $\bar{Q}$.

6. Discussion

Short paragraph: what limits the cell, and what you would change for Part 2.

Technology note

All work is done in the Cadence GPDK 18 nm technology (schematic, layout, Spectre/ADE, DRC/LVS/PEX) at nominal $V_{DD} = 1.2\text{ V}$ unless noted. Characterization above is post-layout unless a subsection states otherwise.